Exp. No: 4 Date:	Measurement and Analysis of Dispersion Compensation in Optical Fiber using OptiSystem

OBJECTIVES

1.To demonstrate dispersion compensation using the Ideal Dispersion Compensation component in OptiSystem.
2.To observe the broadening of a 12.5 ps optical Gaussian pulse after propagation through 10 km of single-mode fiber (SMF).
3.To compensate the accumulated dispersion of 160 ps/nm and verify recovery of the original pulse shape.

SOFTWARE / HARDWARE REQUIRED

•OptiSystem Software

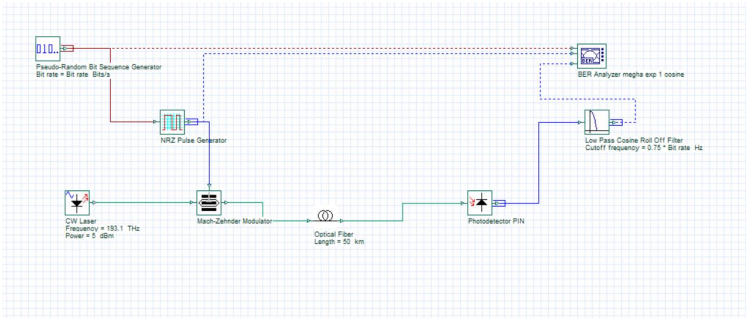
INPUT PARAMETERS

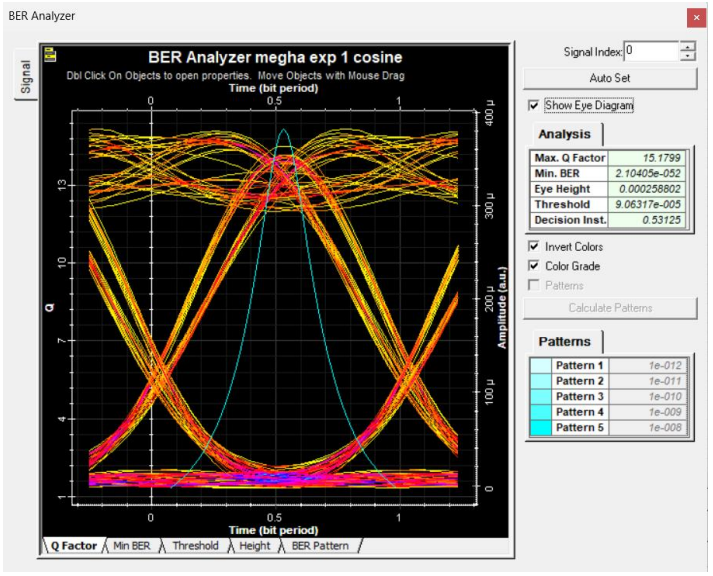
OptiSystem Block	Parameter	Value	Unit / Description
Global Parameters	Bit rate	40	Gb/s
Global Parameters	Pulse width / bit slot	0.5	× bit slot
Optical Gaussian Pulse Generator	Initial pulse width	12.5	ps
Optical Gaussian Pulse Generator	Frequency	193.1	THz
Optical Gaussian Pulse Generator	Power	1	mW
SMF	Length	10	km
SMF	Accumulated dispersion	160	ps/nm
Ideal Dispersion Compensation FBG	Frequency	193.1	THz
Ideal Dispersion Compensation FBG	Bandwidth	125	GHz (~3 × bit rate)
Ideal Dispersion Compensation FBG	Insertion loss	0	dB
Ideal Dispersion Compensation FBG	Depth	100	dB
Ideal Dispersion Compensation FBG	Dispersion	–160	ps/nm
Visualizers / Analyzers	Measurements	Pulse shape, spectrum	Performance analysis

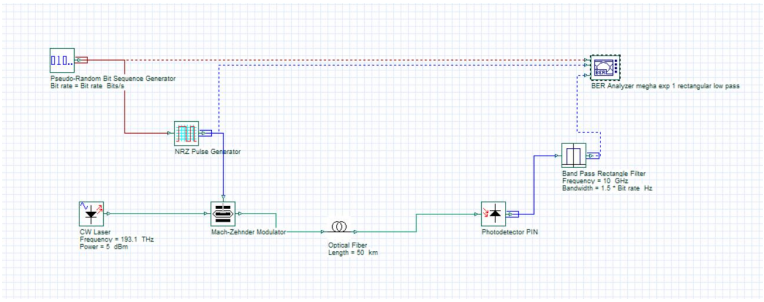
OUTPUT PARAMETERS

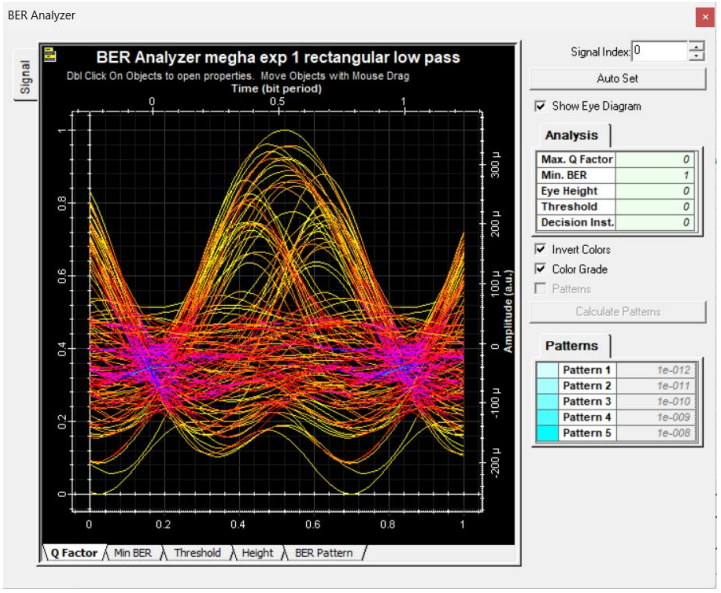
Stage	Observation Point	Result	Figure
Initial pulse	Gaussian pulse	Gaussian pulse, 12.5 ps	Figure 1
After propagation	Output of 10 km SMF	Pulse width increased ≈ 4 times	Figure 2
Compensator sejing	Ideal Dispersion Compensation FBG	Dispersion = –160 ps/nm	Figure 3
After compensation	Compensator output	Original pulse recovered (exact compensation)	Figure 4

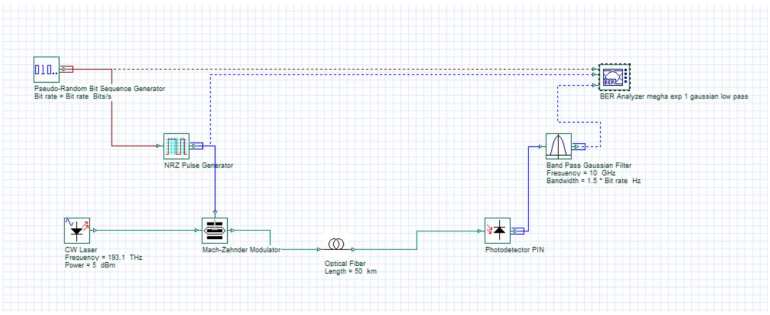
OUTPUT DIAGRAM / GRAPH

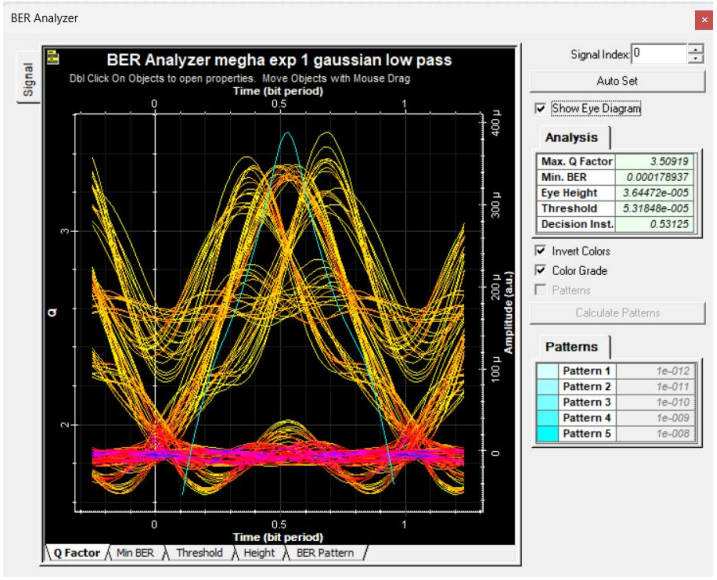


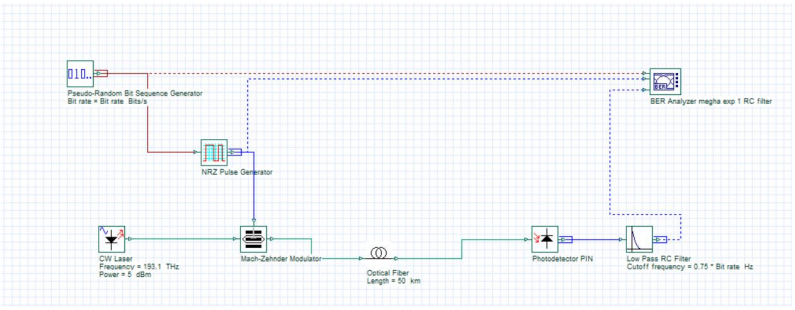


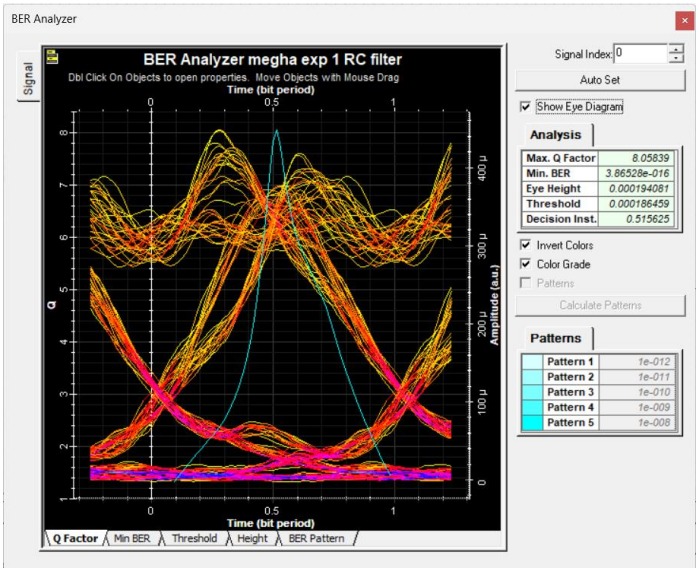





Pseudo-Random Bit Sequence Generator
Bit rate = Bit rate Bits/s
NRZ Pulse Generator
CW Laser
Frequency = 193.1 THz
Power = 5 dBm
Mach-Zehnder Modulator
Optical Fiber
Length = 50 km
Photodetector PIN
Band Pass Gaussian Filter
Frequency = 10 GHz
Bandwidth = 1.5 * Bit rate Hz
BER Analyzer megha exp 1 gaussian low pass


BER Analyzer
BER Analyzer megha exp 1 gaussian low pass
Dbl Click On Objects to open properties. Move Objects with Mouse Drag
Time (bit period)
Amplitude (a.u.)
Q
Signal
Signal Index 0
Auto Set
Show Eye Diagram
Analysis
Max. Q Factor 3.50919
Min. BER 0.000178937
Eye Height 3.64472e-005
Threshold 5.31848e-005
Decision Inst. 0.53125
Invert Colors
Color Grade
Patterns
Calculate Patterns
Patterns
Pattern 1 1e-012
Pattern 2 1e-011
Pattern 3 1e-010
Pattern 4 1e-009
Pattern 5 1e-008
Q Factor
Min BER
Threshold
Height
BER Pattern


Pseudo-Random Bit Sequence Generator
Bit rate = Bit rate Bits/s
NRZ Pulse Generator
CW Laser
Frequency = 193.1 THz
Power = 5 dBm
Mach-Zehnder Modulator
Optical Fiber
Length = 50 km
Photodetector PIN
Low Pass RC Filter
Cutoff frequency = 0.75 * Bit rate Hz
BER Analyzer megha exp 1 RC filter


BER Analyzer
BER Analyzer megha exp 1 RC filter
Dbl Click On Objects to open properties. Move Objects with Mouse Drag
Time (bit period)
Amplitude (a.u.)
Q
Signal
Signal Index 0
Auto Set
Show Eye Diagram
Analysis
Max. Q Factor 8.05839
Min. BER 3.86528e-016
Eye Height 0.000194081
Threshold 0.000186459
Decision Inst. 0.515625
Invert Colors
Color Grade
Patterns
Calculate Patterns
Patterns
Pattern 1 1e-012
Pattern 2 1e-011
Pattern 3 1e-010
Pattern 4 1e-009
Pattern 5 1e-008
Q Factor
Min BER
Threshold
Height
BER Pattern

REFLECTION

I used OptiSystem to launch a 12.5 ps Gaussian pulse at 40 Gb/s through 10 km of single-mode fiber, where the accumulated dispersion of 160 ps/nm broadened the pulse to about four times its original width and would cause inter-symbol interference. I then set the Ideal Dispersion Compensation FBG to –160 ps/nm with a 125 GHz bandwidth centred at 193.1 THz, which restored the original pulse shape and showed that compensation is exact when the compensator value cancels the fiber's dispersion. I learned that the compensator's bandwidth and centre frequency matter as much as its dispersion value, since wrong settings leave the signal clipped or uncompensated. The component was ideal, with no loss or nonlinearity, so a real link would need to account for those effects.

LABORATORY PERFORMANCE ASSESSMENT RUBRIC

Objective	**Assessment Criterion**	**Excellent – 4**	**Good – 3**	**Satisfactory – 2**	**Needs Improvement – 1**	**Weightag e**
Design the OptiSystem layout	Project layout	Builds the complete layout with correct components and connections independently	Minor connection errors corrected	Needs guidance for components or connections	Unable to build the layout	20
Configure pulse, SMF and compensator parameters	Parameter configuration	Sets all parameters (40 Gb/s, 12.5 ps, 193.1 THz, 10 km, −160 ps/nm, 125	Minor parameter errors	Needs guidance for several parameters	Unable to configure parameters	20
Observe pulse broadening in SMF	Pulse observation	Captures and interprets initial and broadened pulses correctly	Minor interpretation errors	Limited observation / needs help	Unable to obtain or interpret pulses	20
Verify compensation using ideal dispersion component	Compensation verification	Verifies recovery of the pulse and explains residual effects of se:ings	Mostly correct verification	Partial verification with assistance	Unable to verify compensation	20
Document findings	Report and conclusion	Complete report with justified conclusion	Minor omissions	Incomplete record / weak conclusion	Report / conclusion inadequate	20

CONCLUSION

Dispersion compensation was successfully demonstrated using the Ideal Dispersion Compensation component in OptiSystem. A 12.5 ps Gaussian pulse broadened approximately four times after 10 km of SMF (accumulated dispersion 160 ps/nm), and sejing the compensator dispersion to –160 ps/nm restored the pulse, achieving an exact compensation of the accumulated dispersion.